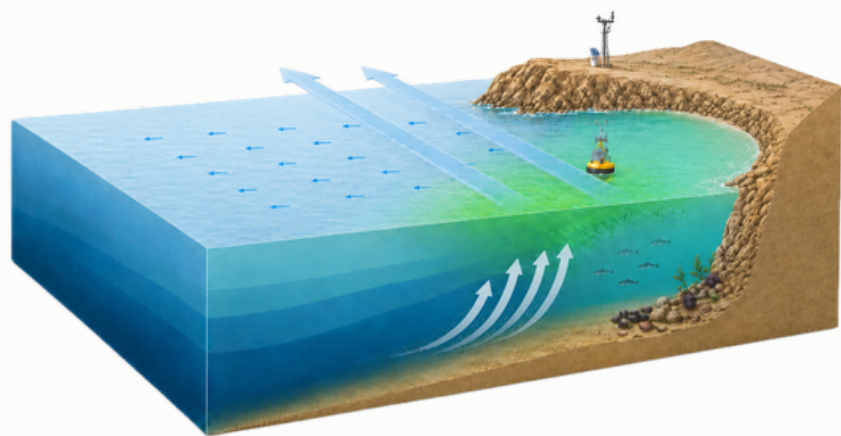


Forecasting chlorophyll in a coastal upwelling system

Daily L4 CHL-a	Copernicus Marine, ~4 km, gap-free
N-central Chile	coastal upwelling system
H01-H10	forecast horizons 1, 3, 5, 7, 10 days

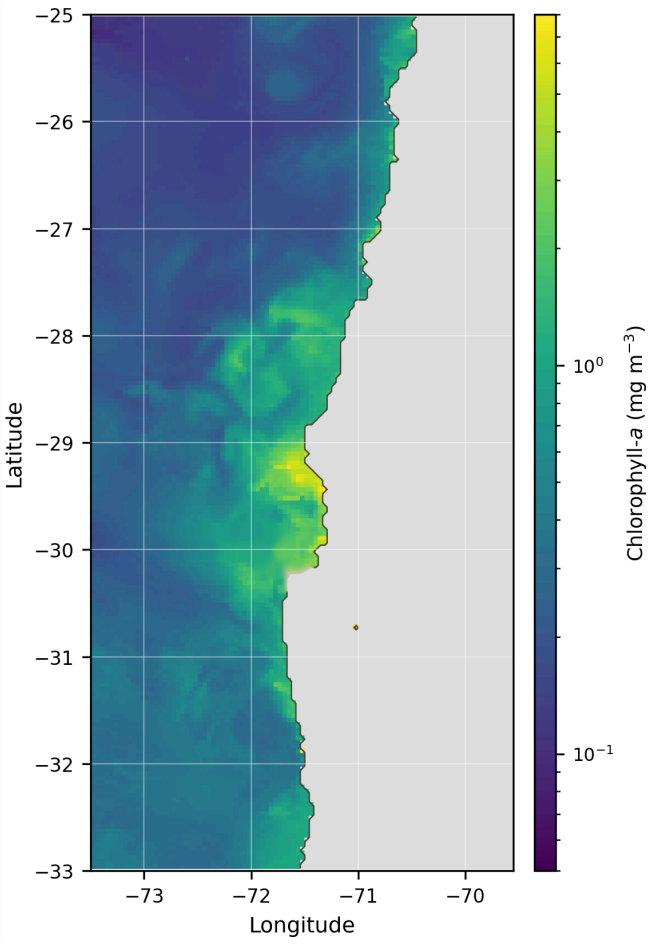
Can we do better than assuming tomorrow looks like today?



Wind drives surface water offshore; cold, nutrient-rich water rises to replace it — that's the chlorophyll source.

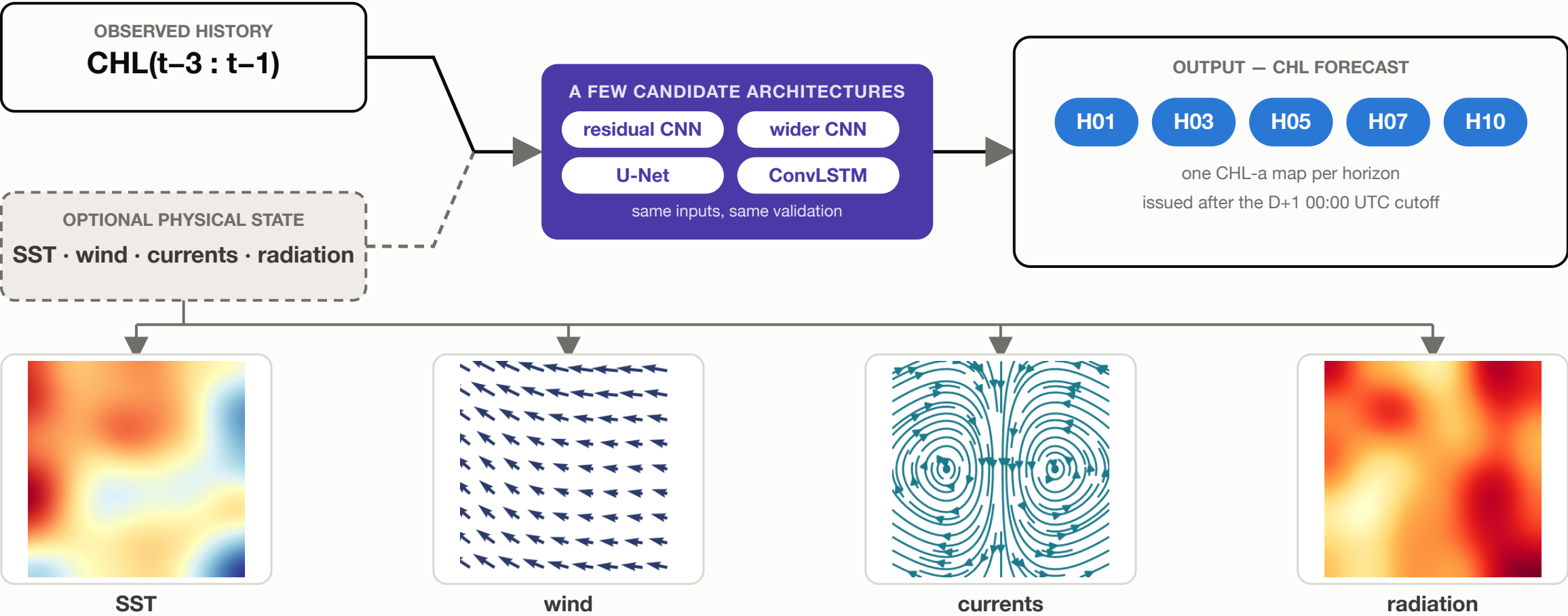


Chile — TARGET region in red



Copernicus Marine L4 chlorophyll-a, one day, over the TARGET domain.

From recent chlorophyll to a forecast



illustrative fields, not a specific date

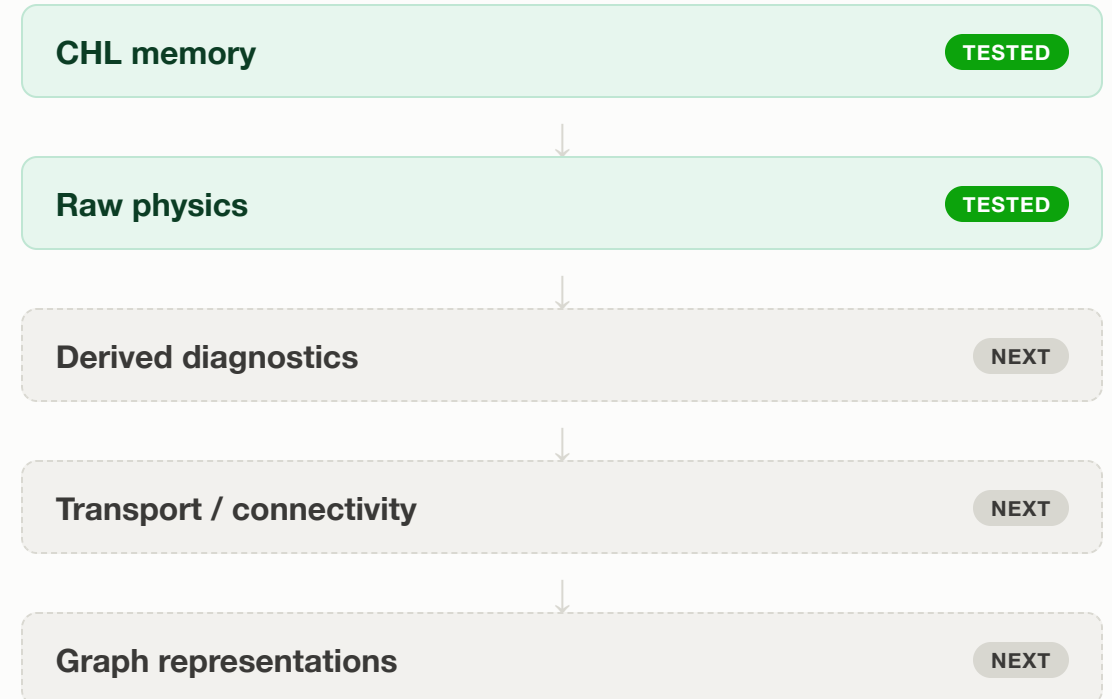
- Blocked temporal validation
- Matched support
- 2023–2025 holdout sealed

What the poster answers

- 1 **Chlorophyll memory** alone is a strong forecast baseline.
- 2 **Raw physical fields** help — but only once you're careful about how you represent them and what you control for.
- 3 Next up is **structured physics**: derived diagnostics, transport and connectivity, graph representations.

Detailed horizon-by-horizon results are on the poster.

RESEARCH LADDER



Poster: *Short-Term Regional Chlorophyll Forecasting* — Matteo Faverio · CEAZA / IPP